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A Machine Learning Project Report [BDS602]

"Real-Time Noise Pollution Monitoring and Sound Classification System"

Submitted in partial fulfillment of the requirements for the award of degree of

Bachelor of Engineering

In

Computer Science And Engineering
[Data Science]

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AY:2025-26

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CERTIFICATE

Certified that the project work entitled **"Real-Time Noise Pollution Monitoring and Sound Classification System"** carried out by **P Sahithi (1AT23CD036), Sneha P (1AT23CD051), Sugitha S (1AT23CD053)** are bonafide students of Department of Computer Science and Engineering [Data Science] , ATRIA IT., Bengaluru, in partial fulfillment for the award of Degree of **Bachelor of Engineering in Computer Science & Engineering [Data Science]** of **Visvesvaraya Technological University, Belagavi**, during the academic year **2025-26**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Ms. Harkiranpreet Kaur Assistant Professor Department of ISE Atria I.T.	**Dr. V Vijayalakshmi** Assistant Professor Department of ISE Atria I.T.	**Dr. Deepak N R** Professor & HOD Department of ISE Atria I.T.	**Dr. Rajesha S** Principal Atria I.T.

DECLARATION

We**, P Sahithi (1AT23CD036) , Sneha P (1AT23CD051) , Sugitha S (1AT23CD053)** students of 6 th semester Bachelor of Engineering, Department of Computer Science and Engineering [Data Science] , Atria Institute of Technology, Bengaluru, would hereby declare that the Project work entitled "**Real-Time Noise Pollution Monitoring and Sound Classification System**" has been carried out by us at Atria Institute of Technology, Bengaluru, and submitted in partial fulfillment of the course requirement for the award of degree of Bachelor of Engineering in Computer Science and Engineering [Data Science] of Visvesvaraya Technological University, Belagavi, during the academic year 2025-26.

We further declare that, to the best of our knowledge and belief, the work embodied in this report has not been submitted to any other university or institution for the award of any other degree.

Place:Bengaluru

Date: 12/05/2026

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ABSTRACT

Noise pollution has become one of the major environmental concerns in urban and industrial areas due to the rapid growth of transportation, construction activities, and public gatherings. Continuous exposure to high noise levels can affect human health, causing stress, hearing loss, and reduced productivity. To address this issue, this project proposes a Real-Time Noise Pollution Monitoring and Sound Classification System using Machine Learning techniques. The system is designed to continuously monitor environmental sounds through a computer microphone and analyze the surrounding acoustic conditions in real time. By combining sound analysis with intelligent classification, the project aims to provide an efficient and automated solution for noise monitoring applications.

The proposed system captures live audio signals and processes them in small time intervals for immediate analysis. Audio preprocessing and feature extraction techniques such as Mel Frequency Cepstral Coefficients (MFCC), frequency spectrum analysis, amplitude detection, and decibel calculation are performed using Librosa and NumPy libraries. A Machine Learning model trained on environmental sound datasets like UrbanSound8K is used to identify different sound categories including traffic noise, sirens, crowd noise, music, human voice, dog bark, and machinery sounds. Based on the calculated decibel level, the system further classifies noise intensity into Very Low, Low, Medium, High, and Very High categories.

The developed application also evaluates the overall environmental condition by categorizing the area into Safe, Moderate Noise, or Noise Pollution Zone. A real-time dashboard created using Streamlit displays the detected sound type, current decibel level, noise classification, and live graphical representation of noise variations. The system is implemented completely in Python using libraries such as Scikit-learn, Pandas, Matplotlib, and Sound Device. This project demonstrates the practical application of Machine Learning and audio signal processing in environmental monitoring and smart city solutions, offering an effective approach for real-time noise pollution assessment and awareness.

ACKNOWLEDGEMENT

We are grateful to our institution, Atria Institute of Technology, for having provided us with the facilities to successfully complete the project (BDS602) on **"Real-Time Noise Pollution Monitoring and Sound Classification System".** We thank **Dr. Rajesha S, Principal** and **Dr. Deepak N R, HOD, ISE** for providing us all the necessary facilities for the successful completion of our project. Deadlines play a very important role in the successful completion of the academic project on time, efficiently and effectively. We take this opportunity to express our deep sense of gratitude to our guide and coordinators **Dr. Vijayalakshmi V, Project Coordinator Assistant Professor**, **Department of ISE** and **Ms. Harkiranpreet Kaur, Assistant Professor, Department of ISE** for their valuable guidance and help throughout the course of the academic project. They have always been patient with us and helped immensely in completing the task on hand. We also thank them for their immense support, guidance, specifications & ideas without which seminar would have been completed without full merit. Last but not least from the **Department of Computer Science and Engineering [Data Science],** teaching and non- teaching staffs for their constant encouragement, support, patience, and endurance shown during the preparation of this report were remarkable. We also thank the management. Finally, we thank our parents and friends for their motivation, morale and material support.

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Chapter 1:

INTRODUCTION

Noise pollution is one of the major environmental problems in modern urban and industrial areas. Increasing traffic, construction activities, industries, public gatherings, and transportation systems produce excessive environmental noise. Continuous exposure to high noise levels can affect human health by causing stress, hearing problems, sleep disturbances, and reduced work efficiency. Therefore, monitoring and controlling environmental noise has become important for maintaining a healthy and safe environment.

Traditional noise monitoring systems mainly measure sound intensity in decibels using sound level meters. However, these systems cannot identify the actual source of the sound or perform intelligent real-time analysis. Manual monitoring methods also require continuous human supervision and are less efficient. Due to these limitations, there is a need for an automated system that can both monitor noise levels and classify environmental sounds accurately.

The proposed project, “Real-Time Noise Pollution Monitoring and Sound Classification System,” uses Machine Learning and audio signal processing techniques to analyze environmental sounds in real time. The system continuously captures live audio through a computer microphone and extracts important audio features such as MFCC, frequency spectrum, amplitude, and decibel levels using Librosa and NumPy libraries. A trained Machine Learning model is then used to classify different sound sources such as traffic noise, sirens, music, shouting, human voice, dog bark, and machinery sounds.

The system also classifies noise intensity into categories such as Very Low, Low, Medium, High, and Very High based on decibel levels. In addition, it determines the environmental condition as Safe, Moderate Noise, or Noise Pollution Zone. A Streamlit dashboard is developed to display real-time results including decibel levels, detected sound type, pollution status, and live noise graphs. This project provides an intelligent, automated, and efficient solution for real-time environmental noise monitoring and sound classification.

Chapter 2:

OBJECTIVES

The main objective of the "Real-Time Noise Pollution Monitoring and Sound Classification System" is to develop an intelligent and automated system capable of monitoring environmental sounds in real time using Machine Learning techniques. The project aims to capture live audio signals through a computer microphone and analyze the surrounding acoustic environment continuously. By using real-time audio processing, the system helps in identifying different environmental sounds and measuring the intensity of noise accurately.

Another important objective of the project is to classify different sound sources such as traffic noise, car horns, sirens, music, human voice, shouting, dog bark, machinery noise, and construction sounds. The system uses audio feature extraction techniques such as MFCC, frequency spectrum analysis, amplitude calculation, and decibel measurement to process audio signals effectively. These extracted features are used by the Machine Learning model to recognize and classify environmental sounds with improved accuracy.

The project also aims to evaluate the level of environmental noise pollution by categorizing sound intensity into Very Low, Low, Medium, High, and Very High levels based on decibel values. In addition, the system determines whether the surrounding environment is Safe, Moderate Noise, or a Noise Pollution Zone. This helps in understanding the severity of environmental noise and supports better monitoring and decision-making.

Another objective of the project is to provide a user-friendly and interactive real-time dashboard using Streamlit. The dashboard displays live decibel levels, detected sound types, pollution status, and graphical representations of noise variations over time. This enables users to monitor environmental conditions continuously and efficiently.

Finally, the project aims to create an effective solution for smart city applications, industrial monitoring, traffic management, hospital zones, schools, and residential areas where environmental noise monitoring is important. The system promotes awareness about noise pollution and demonstrates the practical application of Machine Learning and audio signal processing techniques in real-world environmental monitoring systems.

Chapter 3:

FEATURES

1. Real-Time Sound Monitoring

The system continuously captures environmental audio through a computer microphone and analyzes the sound instantly for continuous monitoring.

2. Automatic Sound Classification

Machine Learning algorithms are used to identify different sound sources such as traffic noise, sirens, music, shouting, and machinery noise.

3. Decibel Level Measurement

The system calculates sound intensity in decibels (dB) and classifies the noise level into categories like Low, Medium, High, and Very High.

4. Noise Pollution Detection

Based on the decibel value, the system determines whether the environment is Safe, Moderate Noise, or a Noise Pollution Zone.

5. Interactive Dashboard

A Streamlit dashboard displays live sound classification results, decibel levels, pollution status, and real-time noise graphs.

6. Real-Time Data Storage

The system stores captured environmental audio data for future analysis and model improvement.

7. Self-Learning Capability

The stored real-time data can be reused to retrain the model and improve sound classification accuracy over time.

Chapter 4:

IMPLEMENTATION

CODE

```
import streamlit as st
import sounddevice as sd
import numpy as np
import joblib
import matplotlib.pyplot as plt
import time
from utils import calculate_db, extract_features, get_pollution_status
# Page Config
st.set_page_config(page_title="Noise Monitor", page_icon="🔊")
# Load Model
try:
    model = joblib.load('noise_model.pkl')
    encoder = joblib.load('encoder.pkl')
except Exception as e:
    st.error(f"Model files not found! Error: {e}")
    st.stop()
st.title("🔊 Real-Time Noise Pollution Monitor")
# Initialize containers
status_placeholder = st.empty()
chart_placeholder = st.empty()
# Audio Settings
SR = 22050
DURATION = 1.5
if 'db_history' not in st.session_state:
    st.session_state.db_history = [0.0] * 20
# Dashboard Logic
st.write("System Status: **Ready to Monitor**")
if st.button('Start Real-Time Tracking'):
    st.write("🔴 Recording Live... (Close browser tab to stop)")
```

```
while True:
    # 1. Capture Audio Live
    rec = sd.rec(int(DURATION * SR), samplerate=SR, channels=1)
    sd.wait()
    audio_data = rec.flatten()

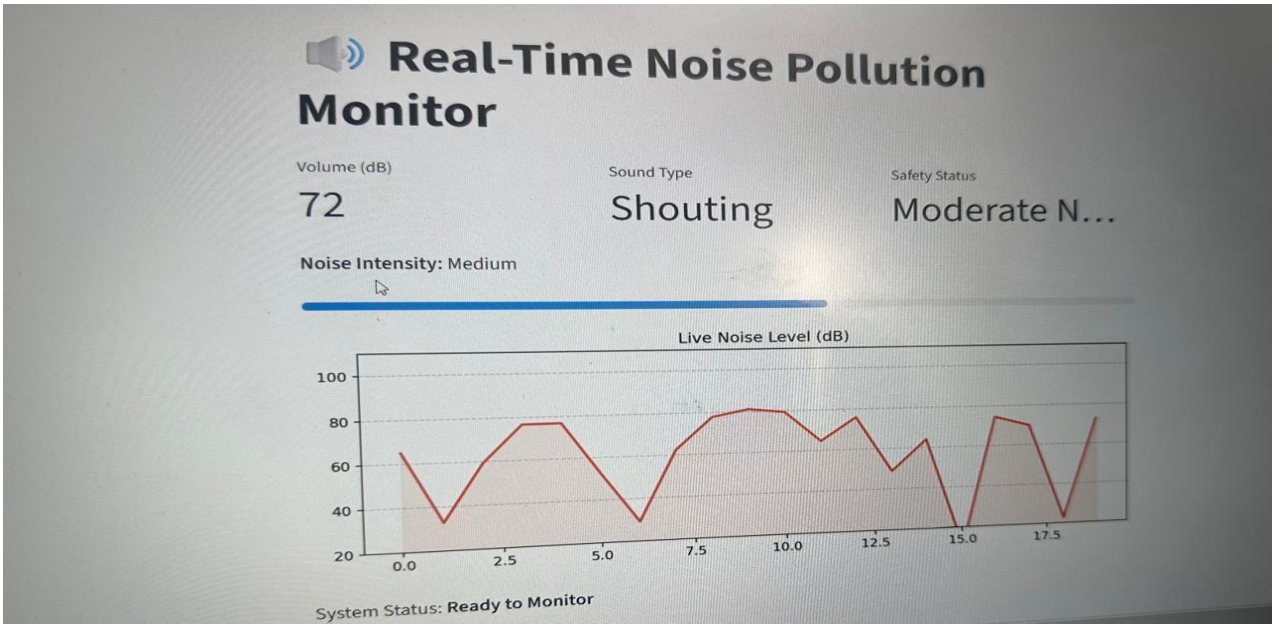
    # 2. Extract Features and Calculate Decibels
    db = float(calculate_db(audio_data)) # Ensure standard Python float

    # If the room is quiet, force 'Silence' to prevent random ML guesses
    if db < 35:
        label = "Silence"
    else:
        features = extract_features(audio_data, SR).reshape(1, -1)
        prediction = model.predict(features)
        label = str(encoder.inverse_transform(prediction)[0])
    status, level_cat = get_pollution_status(db)
    # 3. Update History
    st.session_state.db_history.append(db)
    st.session_state.db_history = st.session_state.db_history[-20:]
    # 4. Update Dashboard UI
    with status_placeholder.container():
        col1, col2, col3 = st.columns(3)
        col1.metric("Volume (dB)", f"{int(db)}")
        col2.metric("Sound Type", label)
        col3.metric("Safety Status", status)
        # Progress calculation with explicit float conversion for Streamlit
        progress_val = min(max((db - 30) / (100 - 30), 0.0), 1.0)
        st.write(f"**Noise Intensity:** {level_cat}")
        st.progress(float(progress_val)) # FIX: Explicit float conversion
```

```
# 5. Plot Live Graph
        fig, ax = plt.subplots(figsize=(10, 3))
        ax.plot(st.session_state.db_history, color='#ff4b4b', linewidth=2)
        ax.fill_between(range(len(st.session_state.db_history)),
st.session_state.db_history, color='#ff4b4b', alpha=0.1)
ax.set_ylim([20, 110])
        ax.set_title("Live Noise Level (dB)")
        ax.grid(axis='y', linestyle='--', alpha=0.7)
        chart_placeholder.pyplot(fig)
        plt.close(fig)
        time.sleep(0.1)
```

Chapter 5:

RESULTS AND DISCUSSION



The developed Real-Time Noise Pollution Monitoring System successfully captures and analyzes environmental sound in real time using Machine Learning techniques. The system continuously records live audio through the microphone and calculates the corresponding decibel (dB) level. In the observed output, the detected sound intensity is around 72 dB, which is classified as a Medium noise level. The system also identifies the detected sound type as "Shouting" and categorizes the environmental condition under Moderate Noise status.

The live noise graph displayed on the dashboard shows continuous fluctuations in decibel levels over time, indicating dynamic environmental sound variations. This demonstrates the capability of the system to monitor changing acoustic conditions continuously and provide instant updates. The integration of real-time sound classification and noise level detection improves monitoring accuracy compared to traditional sound level measurement systems.

The Streamlit dashboard provides an interactive and user-friendly interface for visualizing real-time monitoring results, including sound type, decibel level, noise intensity, and graphical noise analysis. Overall, the system effectively performs environmental sound classification and noise pollution analysis, making it suitable for applications such as smart city monitoring, traffic analysis, industrial monitoring, and public safety systems.

Chapter 6:

CONCLUSION

The “Real-Time Noise Pollution Monitoring and Sound Classification System” successfully demonstrates the use of Machine Learning and audio signal processing techniques for intelligent environmental noise monitoring. The system captures live environmental sounds through a computer microphone and analyzes them in real time. Using audio preprocessing and feature extraction methods such as MFCC, frequency spectrum analysis, amplitude calculation, and decibel measurement, the system processes audio signals efficiently and identifies different environmental sound sources. The Machine Learning model helps classify sounds such as traffic noise, sirens, music, human voice, shouting, dog bark, and machinery noise with good accuracy.

The proposed system performs both noise level classification and sound type classification simultaneously. Based on calculated decibel levels, the system categorizes environmental noise into Very Low, Low, Medium, High, and Very High levels. It also determines the environmental condition as Safe, Moderate Noise, or Noise Pollution Zone. This real-time analysis helps users understand the severity of environmental noise and supports effective monitoring and public awareness.

The project also provides an interactive Streamlit dashboard that displays live decibel readings, detected sound types, pollution status, and graphical representations of noise variations. Compared to traditional monitoring systems, the proposed system provides automation, intelligent analysis, improved accuracy, and faster processing. The system can be applied in smart cities, industries, schools, hospitals, traffic monitoring areas, and residential zones where continuous environmental monitoring is important.

In future, the system can be enhanced by using advanced Deep Learning models such as CNN and RNN for higher sound classification accuracy. Additional features such as IoT integration, cloud storage, mobile application support, automatic alert systems, and GPS-based monitoring can also be implemented. Overall, the proposed system provides an effective, reliable, and scalable solution for real-time noise pollution monitoring and intelligent environmental sound analysis.

Chapter 6:

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Chapter 6:

CONCLUSION